支持不使用 ASMLib 的 ASM 数据

SnapManager Oracle

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支持不使用 ASMLib 的 ASM 数据□

默□情况下， SnapManager 支持不使用 ASMLib 的 ASM 。基本要求是，必□□用于 ASM 磁□□的□□□行分区。

如果未安装 ASMLib ， □在□行以下操作□，与 ASM 磁□□相□的□□□限将更改□ root ： disk ： disk ：

- 重新□□主机
- 使用基于卷的 SnapRestore （ VBSR ）从主存□□原数据□
- 从二□存□□原数据□

□可以通□□ SMO.conf 中的 oracleasma.support.without .asmlib 配置□量分配 true 来□置□当的□□□限。□当在主机中添加或□除新□□□，系□都会在 initasmdisks 文件中添加或□除与 ASM 磁□□相□的□□。initasmdisks 文件位于 /etc/initasmdisks 。

例如，如果□置 oracleasma.support.without .asmlib=true ， 然后□行□□挂□，□会将新□□添加到 initasmdisks 中。重新□□主机后，□□脚本会保留□□□限和所有□。

> oracleasma.support.without 。 asmlib 的默□□□ false 。

- 相□信息 *

支持的分区□□

支持的脚本

使用 asmmain.sh 和 asmquerydisk.sh 脚本可以更改网格用□，□和用□，所有□些脚本都用于□□ ASM 磁□。必□始□从根□行□些脚本。

asmmain.sh 是从任何添加或□除□□的操作中□用的主脚本文件。asmmain.sh 脚本会在内部□用□一个脚本，□脚本需要从具有 Oracle 网格凭据的根目□□行。此脚本将□□ ASM 磁□□的□□，然后在 initasmdisk 文件中添加□些条目，并添加□些□□的□限和所有□。□可以根据□的□境以及□用于匹配 /dev/mapper/* P1 的正□表□式更改此文件的□限和所有□。

asmquerydisk.sh 脚本用于□□磁□列表，□列表用于□建 ASM 磁□□。□必□根据□的配置□ ORACLE_BASE ， ORACLE_HOME 和 ORACLE_SID 分配□。

□些脚本位于 /opt/netapp/smO/plugins/examples/noasmlib 。但是，在主机上□□ SnapManager for Oracle 服□器之前，必□将□些脚本移至 /opt/netapp/smO/plugins/noasmlib 。

使用脚本支持不使用 ASMLib 的 ASM 数据□的限制

□必□了解在不使用 ASMLib 的情况下使用脚本支持 ASM 数据□的某些限制。

- □些脚本可□任何内核版本提供□用解决方案，但前提是未安装 ASMLib 。
- 脚本的□限□置方式必□使 root ， grid ， oracle 或等效用□能□□□□些脚本。
- □些脚本不支持从二□位置□行□原。

部署和□行脚本

□可以部署和□行 asmmain.sh 和 asmquerydisk.sh 脚本，以支持 ASM 数据□，而无需 ASMLib 。

□些脚本不遵循□脚本或后脚本□法，□用 intitasmdisks □会□用工作流。□可以在脚本中更改与配置□置相□的任何内容。建□通□□行快速□□行来□□脚本中的所有内容是否均按□期工作。

> □些脚本不会因故障而□系□造成□害，也不会影□系□。□行□些脚本可更新与 ASM 相□的磁□以□得□当的□限和所有□，从而使□些磁□始□□于 ASM □例控制之下。

1. 使用已分区的磁□□建 ASM 磁□□。
2. 在磁□□上□建 Oracle 数据□。
3. 停止□用于 Oracle 服□器的 SnapManager 。

   > 在 RAC □境中，□需要□所有 RAC □点□行此□□。

4. 修改 smo.conf 以包含以下参数：

   a. oracleasma.support.without .asmlib = true
   b. oracleasma.support.without .asmlib.ownership = true
   c. oracleasm.support.without.asmlib.username = ASM □例□境的用□名
   d. oracleasm.support.without.asmlib.groupname = ASM □例□境的□名称

   □些修改□□□□路径□置□限，□意味着，将□□ dm-* □□□置□限，而不是分区□□。

5. 修改 /opt/netapp/smO/plugins/examples/noasmlib 中提供的□件脚本，以便在□些脚本中包含□的配置□置。
6. 在主机上□□ SnapManager for Oracle 服□器之前，□将脚本□制到 /opt/netapp/smO/plugins/noasmlib 。
7. □航到 /opt/netapp/smo 目□，然后□行以下脚本□行□□行： sh plugins/noasmlib/asmmain.sh

   此□将□建 etc/initasmdisks 文件，□文件是所使用的主文件。

   □可以□□ etc/initasmdisks 文件包含与已配置的 ASM 数据□相□的所有□□，例如：

```
chown -R grid:oinstall /dev/mapper/360a98000316b61396c3f394645776863p1
         chmod 777 /dev/mapper/360a98000316b61396c3f394645776863p1
         chown -R grid:oinstall
/dev/mapper/360a980003754322f7a2b433469714239p1
         chmod 777 /dev/mapper/360a980003754322f7a2b433469714239p1
         chown -R grid:oinstall
/dev/mapper/360a980003754322f7a2b433469714241p1
         chmod 777 /dev/mapper/360a980003754322f7a2b433469714241p1
         chown -R grid:oinstall
/dev/mapper/360a980003754322f7a2b433469714243p1
         chmod 777 /dev/mapper/360a980003754322f7a2b433469714243p1
```

8. □□□用于 Oracle 服□器的 SnapManager 。
9. 通□将以下内容添加到 SnapDrive.conf file.disconnect -luns-before-vbsr=on 来配置 SnapDrive for UNIX
10. 重新□□□用于 UNIX 的 SnapDrive 服□器。

> 在 RAC □境中，□需要□所有 RAC □点□行□□ 3 到□□ 10 。

□建的 /etc/initasmdisks 文件必□从其中一个□□脚本或 rc3.d 中新定□的脚本□行在 oracleha 服□□□之前，□始□□行 /etc/initasmdisks 文件。

示例

```
# ls -ltr *ohasd*
          lrwxrwxrwx 1 root root 17 Aug  7 02:34 S96ohasd ->
/etc/init.d/ohasd
          lrwxrwxrwx 1 root root 17 Aug  7 02:34 K15ohasd ->
/etc/init.d/ohasd
```

在以下示例中，默□情况下， sh -x/etc/initasmdisks 不可用，□需要将其附加□ ohasd 脚本中函数 `stt_stack（）` 的第一行：

```
start_stack()
{
sh -x /etc/initasmdisks
# see init.ohasd.sbs for a full rationale case $PLATFORM in Linux
}
```

支持不使用 **ASMLib** 的 **Oracle RAC ASM** 数据□

如果使用的是 Oracle RAC 数据□，□□当在主 RAC □点中□行操作□，都必□使用 initasmdisks 文件更新 RAC □点。

如果从主□点登□到 RAC □点不需要身□□□，□ asmmain.sh 会□所有 RAC □点□行 initasmdisks 的安全□制（ SCP ）。□当□行□原□，都会□用主□点的 initasmdisks 文件，并且可以更新 asmmain.sh 脚本，以便在所有 RAC □点中□用相同的脚本。

□建的 /etc/initasmdisks 文件必□从其中一个□□脚本或 rc3.d 中新定□的脚本□行在 oracleha 服□□□之前，□始□□行 /etc/initasmdisks 文件。

支持不使用 **ASMLib** 的 **Oracle 10g ASM** 数据□

如果使用的是 Oracle 10g ，□ asmcmd 命令不可用于列出磁□。□可以使用 SQL □□□取磁□列表。

disk_list.sql 脚本包含在示例目□中提供的□有脚本中，用于支持 SQL □□。□行 theasmquerydisk.sh 脚本□，必□手□□行 disk_list.sql 脚本。在 asmquerydisk.sh 文件中添加了示例脚本行以及注□。此文件可以放置在

/home/grid 位置或□□□的其他位置。

支持 ASM 数据□而不使用 ASMLib 的示例脚本

示例脚本位于 SnapManager for Oracle 安装目□的 plugins/examples/noasmlib 目□中。

asmmain.sh

```
#!/bin/bash
griduser=grid
gridgroup=oinstall

# Run the script which takes the disklist from the asmcmd
# use appropriate user , here grid user is being used to run
# asmcmd command.
su -c "plugins/noasmlib/asmdiskquery.sh" -s /bin/sh grid
cat /home/grid/disklist

# Construct the final file as .bak file with propre inputs
awk -v guser=$griduser -v ggroup=$gridgroup '/^\/dev\/mapper/ { print
"chown -R "guser":"ggroup" "$1; print "chmod 777 " $1; }'
/home/grid/disklist > /etc/initasmdisks.bak

# move the bak file to the actual file.
mv /etc/initasmdisks.bak /etc/initasmdisks

# Set full full permission for this file to be called while rebooting and
restore
chmod 777 /etc/initasmdisks

# If the /etc/initasmdisks needs to be updated in all the RAC nodes
# or /etc/initasmdisks script has to be executed in the RAC nodes, then
the following
# section needs to be uncommented and used.
#
# Note: To do scp or running scripts in remote RAC node via ssh, it needs
password less login
# for root user with ssh keys shared between the two nodes.
#
# The following 2 lines are used for updating the file in the RAC nodes:
# scp /etc/initasmdisks root@racnode1:/etc/initasmdisks
# scp /etc/initasmdisks root@racnode2:/etc/initasmdisks
#
# In order to execute the /etc/initasmdisks in other RAC nodes
# The following must be added to the master RAC node /etc/initasmdisks
file
```

```
# from the asmmain.sh script itself. The above scp transfer will make sure
# the permissions and mode for the disk list contents are transferred to
the other RAC nodes
# so now appending any command in the /etc/initasmsdisks will be retained
only in the master RAC node.
# The following lines will add entries to the /etc/initasmsdisks file in
master RAC node only. When this script is executed
#  master RAC node, /etc/initasmdisks in all the RAC nodes will be
executed.
# echo 'ssh racnode1 /etc/initasmdisks' >> /etc/initasmdisks
# echo 'ssh racnode2 /etc/initasmdisks' >> /etc/initasmdisks
```

asmquerydisk.sh

```
#!/bin/bash
export  ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=/u01/app/grid/product/11.2.0.3/grid
export ORACLE_SID=+ASM
export  PATH=$ORACLE_HOME/bin:$PATH

# Get the Disk List and save this in a file called dglist.
asmcmd lsdsk  > /home/grid/disklist

# In oracle 10g the above used command 'asmcmd' is not available so use
SQL
# query can be used to take the disk list. Need to uncomment the following
# line and comment the above incase oracle 10g is being in use.
# The disk_list.sql script is availbe in this noasmlib examples folder
itself
# which can be modified as per customer needs.
# sqlplus "/as sysdba" @/home/grid/disk_list.sql > /home/grid/disklist
```

disk_list.sql

```
# su - oracle
-bash-4.1$ cat disk_list.sql
select path from v$asm_disk;
exit
-bash-4.1$
```

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